

[Total No. of Questions - 9] [Total No. of Printed Pages - 3]

May-25-1104

MCA-6206(iii) (Information Security)

MCA 2nd (NEP)

Time : 3 Hours

Max. Marks : 60

The candidates shall limit their answers precisely within the answer-book (40 pages) issued to them and no supplementary/continuation sheet will be issued.

Note : Attempt five questions in all. Question no. 9 in section E is compulsory. Rest attempt one question each from the sections A, B, C & D.

SECTION - A

1. (a) Describe how hashing functions used in security applications. (6)
(b) How does public-key encryption work? Explain with a suitable example. (6)

OR

2. (a) Explain digital signatures and their role in data integrity. (6)
(b) Explain the basic principles of computer security and their significance. (6)

SECTION - B

3. (a) What are access rights? How are they assigned to subjects and objects? (6)
(b) Discuss the working of discretionary access control with an example. (6)

OR

2

MCA-6206(iii)

4. (a) Explain different authentication methods and compare their security levels. (6)
(b) Discuss the working of token-based authentication with a suitable example. (6)

SECTION - C

5. (a) Explain the role of DBMS in database security. How are relational databases managed securely? (6)
(b) Discuss different database access control mechanisms and compare their effectiveness. (6)

OR

6. (a) What is the significance of user authentication in database security? (6)
(b) Discuss the impact of DoS attacks on government organizations. (6)

SECTION - D

7. (a) Discuss some common physical security prevention measures. (6)
(b) How do natural disasters and environmental threats impact infrastructure security? (6)

OR

8. (a) How does access control enhance physical security? Provide suitable examples. (6)
(b) What are the key components of a corporate physical security policy? (6)

[P.T.O.]